

NILOY JANA

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EDUCATION

St Thomas College of Engineering and Technology - B.Tech Computer Science and Engineering [2023-2027]
South Point High School - (CBSE) X-94% | (CBSE) XII - 74% [physics, chemistry, mathematics, biology] [2008-2021]

EXPERIENCE

AI/ML Engineer Intern | KreupAi Technologies LLC |April 2026 - July 2026

- Preprocessed and transformed datasets to support machine learning workflows and scalable data pipelines.
- Built and tested ML/DL models for regression, classification, clustering, and predictive analytics using Python frameworks.
- Applied feature engineering and hyperparameter tuning to improve model accuracy and performance.
- Evaluated models using Accuracy, Precision, Recall, F1-Score, and RMSE while documenting workflows for reproducibility and deployment.

Artificial Intelligence Intern | Pratinik Infotech |March 2026 - May 2026

- Developed a machine learning-based predictive maintenance model achieving high failure prediction accuracy for equipment forecasting and downtime reduction.
- Preprocessed and analyzed sensor data to train optimized predictive models for failure detection and anomaly analysis.
- Implemented a real-time monitoring and alerting pipeline for proactive maintenance recommendations.
- Built an interactive dashboard to visualize equipment health metrics and maintenance insights.

PROJECTS

ARIA - Multi-Agent AI Financial Advisory | Python | RAG (FAISS vector store)|REST API | SQLite

[GitHub](#)

- Developed ARIA, a full-stack, privacy-first AI financial advisory web application (FastAPI + Vanilla JS) integrating a multi-agent LLM reasoning system (Ollama/Groq) to deliver personalized, tool-driven financial planning across 8 domains — including planning, tax optimization, portfolio analysis, and couples' finance management.
- Engineered a coordinated multi-agent orchestration pipeline (Auditor + Market Specialist + Manager agents) with RAG-powered document analysis and a dual AI-provider architecture supporting fully offline local inference via Ollama — eliminating cloud data dependency unlike conventional AI finance tools.

Opsira - AI Eye Disease Detection System | Python | TensorFlow/Keras | OpenCV | Flask

[GitHub](#)

- Developed a computer vision-based eye health screening system detecting Cataract, Red Eye (Conjunctivitis), Jaundice, and Healthy eyes from images using CNN-based deep learning and image processing techniques.
- Implemented Explainable AI (Grad-CAM heatmaps) to visualize regions influencing predictions and built a fallback rule-based analysis extracting physiological metrics (redness, yellowness, cloudiness) for risk assessment.

Sign Language detector - Real Time Hand Gesture Recognition System |Python| OpenCV |Scikit-Learn

[GitHub](#)

- Built a real-time hand gesture recognition system using a webcam to detect hands, classify gestures, and display prediction confidence using Python, OpenCV, MediaPipe, and machine learning for touch-free interaction.

TECHNICAL SKILLS

Language - Python , JavaScript

Framework - TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, MediaPine, OpenCV, LangChain, FastAPI, Flask, Node.js

AI/ML - Generative AI, RAG (FAISS), Prompt Engineering, LLM Integration, LLM Agent Design, Explainable AI

Web - HTML, CSS, REST APIs

Databases - MySQL, SQLite

Tools - Git, GitHub

CERTIFICATIONS

- Infosys Certified in Deep Learning** - Demonstrating proficiency in AI/ML concepts and practical model development [April 2026]
- IBM AI Engineering Professional Certificate** - Comprehensive training in Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Transformers, and AI Agent Development [May 2026]

ACHIEVEMENTS

- Achieved 1st Place in Inter-College Hackathon by developing a competitive solution under time constraintsFinalist
- AI Hackathon - Indian Institute of Technology Delhi: Ranked Top 8 out of 1400+ participants; built a hybrid LLM + rule-based AI agent for an Among Us simulation with BFS navigation, chat analysis, and strategic voting logic.
- Led the organization and execution of a College-Level Coding Competition, managing participants and event coordination